



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2024 - 2025)

LEADER & ACHIEVER COURSE

PHASE : MLB, MLC, MLK, MLM, MLN, MLP, MLQ, MLR, MLS, MLV, MAZE, MAZF, MAZG, MAZH, MAZI, MAZJ, MAZM, MAZP, MAZO, MAZR, MAZS, MAZT, MAZU, MAZV, MAAZ, MAAW, SPARKPRO (SET - R2)

TARGET : PRE MEDICAL 2025

Test Type : MAJOR

Test Pattern : NEET (UG)

TEST DATE : 23-04-2025

ANSWER KEY : SET # R2

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	3	4	2	2	2	4	3	4	3	2	3	4	2	2	4	2	2	2	3	1	3	1	4	3	2	2	1	1	4	2
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	2	4	3	3	1	2	2	3	1	2	1	1	1	2	2	4	3	3	1	2	4	4	2	4	3	1	1	3	2	1
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	2	2	3	3	3	2	4	2	1	3	4	1	2	1	1	2	2	4	1	2	1	3	2	2	2	3	4	4	3	4
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	3	3	3	3	1	2	1	3	2	1	3	2	4	1	3	3	3	2	2	3	3	1	1	1	1	4	3	3	2	4
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	2	4	2	4	1	3	4	4	4	3	3	2	4	3	2	2	2	4	1	2	2	3	2	3	4	3	4	3	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	2	1	3	4	1	4	2	3	4	3	3	2	4	2	4	1	4	2	1	2	1	2	3	2	1	4	2	1	4	2

HINT - SHEET : SET# R2

1. Ans (3)

$$\text{Basic strength} \propto \frac{+M}{-M} \propto \frac{+H}{-H} \propto \frac{+I}{-I}$$

2. Ans (4)

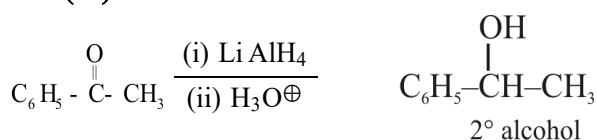


Hybridization of carbon atom is changed from sp^3 to sp^2 , therefore the above reaction is an elimination reaction.

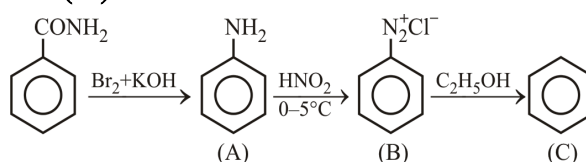
5. Ans (2)

E2 will be favoured in option 1 whereas S_N2 will be favoured in option 2.

10. Ans (2)



15. Ans (4)



31. Ans (2)

Due to polarisation as compound is BeCl_2

45. Ans (2)

Inner orbital complex means in presence of strong field ligand.

46. Ans (4)

NCERT Pg. No. # 59

48. Ans (3)

NCERT Pg. No. # 64, 65, 66

49. Ans (1)

NCERT Pg. No. # 71

51. Ans (4)

NCERT Pg. No. # 71

52. **Ans (4)**
NCERT XI Page No. 26, 27, 28
53. **Ans (2)**
NCERT XI, Page No. 10
54. **Ans (4)**
NCERT XI, Page No. 30, 32
55. **Ans (3)**
NCERT XI, Page No. 24
56. **Ans (1)**
NCERT XI, Page No. 17
57. **Ans (1)**
NCERT XI, Page No. 30, 32
58. **Ans (3)**
NCERT XI, Page No. 4, 7, 23, 24
59. **Ans (2)**
NCERT Pg # 117
60. **Ans (1)**
NCERT Pg # 175-177
61. **Ans (2)**
NCERT Pg # 150
62. **Ans (2)**
NCERT Pg # 146
63. **Ans (3)**
NCERT Pg # 158
64. **Ans (3)**
NCERT Pg # 156
65. **Ans (3)**
NCERT Pg # 162
66. **Ans (2)**
NCERT Pg # 146
67. **Ans (4)**
NCERT Page No. 197
68. **Ans (2)**
NCERT Page No. 192
69. **Ans (1)**
NCERT Page No. 200
70. **Ans (3)**
NCERT Page No. 207

71. **Ans (4)**
NCERT Page No. 212
72. **Ans (1)**
NCERT Page No. 217
73. **Ans (2)**
NCERT Page No. 221
74. **Ans (1)**
NCERT Page No. 222
75. **Ans (1)**
NCERT-XII, Pg#69
76. **Ans (2)**
NCERT-XII, Pg#61
77. **Ans (2)**
NCERT-XII, Pg#60
78. **Ans (4)**
NCERT-XII, Pg#76
79. **Ans (1)**
NCERT-XII, Pg#55,56,57,59
80. **Ans (2)**
NCERT XII Pg # 157
81. **Ans (1)**
NCERT XII Pg # 152, 153
82. **Ans (3)**
NCERT Pg # 81
83. **Ans (2)**
NCERT Pg # 81
84. **Ans (2)**
NCERT Pg # 81
85. **Ans (2)**
NCERT Pg # 81
86. **Ans (3)**
NCERT Pg # 105
87. **Ans (4)**
NCERTG Pg # 94
88. **Ans (4)**
NCERT Pg # 9
89. **Ans (3)**
NCERT Pg # 8, 19, 20, 22

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|---|--|
| <p>90. Ans (4)
NCERT Pg # 12, 13</p> <p>91. Ans (3)
NCERT-XI, Page No.: 91</p> <p>92. Ans (3)
NCERT-XI, Page No.: 125</p> <p>93. Ans (3)
NCERT-XI, Page No.: 101</p> <p>94. Ans (3)
NCERT-XI, Page No.: 97, 98</p> <p>95. Ans (1)
NCERT-XII Pg#182(E), 199(H)</p> <p>96. Ans (2)
NCERT-XII Pg#170(E), 219(H)</p> <p>97. Ans (1)
NCERT-XII Pg#106(E), 106(H)</p> <p>98. Ans (3)
NCERT-XII Pg#111,112 (E), 111,112(H)</p> <p>99. Ans (2)
NCERT-XII Pg#209</p> <p>100. Ans (1)
NCERT-XII Pg#39</p> <p>101. Ans (3)
NCERT-XII Pg#41</p> <p>102. Ans (2)
NCERT-XII Pg#49</p> <p>103. Ans (4)
NCERT-XII Pg#47</p> <p>104. Ans (1)
NCERT-XII Pg # 82</p> <p>105. Ans (3)
NCERT Pg. No. 203</p> <p>106. Ans (3)
NCERT Page No. # 199/200</p> <p>107. Ans (3)
NCERT Pg. No. 194/195</p> <p>108. Ans (2)
NCERT Pg. No. 184</p> | <p>109. Ans (2)
NCERT Pg. No. 187</p> <p>110. Ans (3)
NCERT Pg. No. 189/190</p> <p>111. Ans (3)
NCERT Pg. No. # 208</p> <p>112. Ans (1)
NCERT Page No. #241</p> <p>113. Ans (1)
NCERT Page No. #242</p> <p>114. Ans (1)
NCERT Page No. #244</p> <p>115. Ans (1)
NCERT Page No. #242, 243, 244</p> <p>116. Ans (4)
NCERT Page No. #247</p> <p>117. Ans (3)
NCERT Pg. No. # 232</p> <p>118. Ans (3)
NCERT Pg. No. # 236</p> <p>119. Ans (2)
NCERT Pg. No. # 234</p> <p>120. Ans (4)
NCERT Pg. No. # 219</p> <p>121. Ans (2)
NCERT Pg. No. # 227</p> <p>122. Ans (2)
NCERT Page No.: 141(E), 154(H)</p> <p>123. Ans (4)
NCERT Page No.: 131(E)</p> <p>124. Ans (2)
NCERT Page No.: 121</p> <p>125. Ans (4)
NCERT Page No.: 125</p> <p>126. Ans (1)
NCERT Page No.: 44</p> <p>127. Ans (3)
NCERT Page No.: 47, Para.: 3.4</p> |
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128. Ans (4)

NCERT Page No.: 28-30

129. Ans (4)

NCERT Page No.: 111

130. Ans (4)

NCERT Page No.: 114

131. Ans (3)

NCERT Page No.: 116

132. Ans (3)

NCERT Page No.: 27

133. Ans (2)

NCERT Page No.: 44

134. Ans (4)

NCERT Page No.: 37

135. Ans (3)

NCERT Page No.: 120, 121

136. Ans (2)

Applying law of energy conservation

$$T.KE = T. PE$$

$$\frac{1}{2}mU^2 = \frac{KQq}{r} \quad \dots(i)$$

$$\frac{1}{2}m\left(\frac{U}{4}\right)^2 = \frac{KQq}{r'} \quad \dots(ii)$$

$$(i) \div (ii)$$

$$16 = \frac{r'}{r} \Rightarrow r' = 16r$$

137. Ans (2)

$$U_{\text{dipole}} = -\vec{P} \cdot \vec{E} = -PE \cos \theta$$

$$= -P \left(\frac{2\sigma}{\epsilon_0} \right) \cos(0^\circ) = \frac{-2\sigma P}{\epsilon_0}$$

138. Ans (2)

When a dielectric K is introduced in a parallel plate condenser its capacity becomes K times.

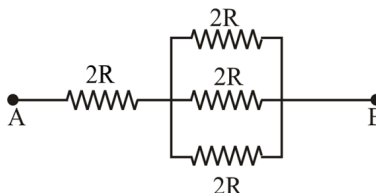
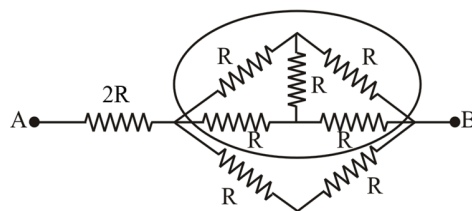
$$\text{Hence } C' = 4C_0.$$

$$\text{Energy stored } W_0 = \frac{q^2}{2C_0}$$

$$\therefore W' = \frac{q^2}{2C'} = \frac{q^2}{2 \times 4C_0} \Rightarrow W' = \frac{W_0}{4}$$

139. Ans (4)

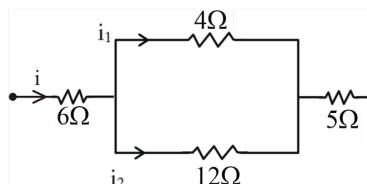
Balanced W.S.B.



$$R_{AB} = 2R + \frac{2R}{3} = \frac{8R}{3}$$

140. Ans (1)

$$P_{12} = 12W$$



$$i_2 = \frac{4}{16} \times i = \frac{1}{4}i$$

$$\therefore P_{12} = i_2^2 \times 12$$

$$12 = \left(\frac{i}{4}\right)^2 \times 12$$

$$i^2 = 16$$

$$i = 4A$$

$$\text{so } P_3 = i^2 \times 6 = (4)^2 \times 6 = 96W$$

142. Ans (2)

$$W = MB(1 - \cos 60^\circ) = 25J$$

143. Ans (3)

$$\frac{E}{B} = C$$

$$B = \frac{E}{C} = \frac{1260}{3 \times 10^8} = 4.2 \times 10^{-6} \text{ Wb/m}^2$$

144. Ans (2)

For equivalent focal length,

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3} = \frac{1}{(+20)} + \frac{1}{(-20)} + \frac{1}{(-30)}$$

$$\Rightarrow \frac{1}{f} = -\frac{1}{30} \Rightarrow f = -30\text{cm}$$

Since, power i.e., P (in D)

$$= \frac{1}{f(\text{in m})} = -\frac{100}{30} = -\frac{10}{3} = -3.33 \text{ D}$$

145. Ans (3)

At point A,

$$I_A = I + 9I + 2\sqrt{I}\sqrt{9I} \cos \pi$$

$$= 4I$$

At point B,

$$I_B = I + 9I + 2\sqrt{I}\sqrt{9I} \cos 2\pi$$

$$= 16I$$

$$\therefore I_B - I_A = 16I - 4I = 12I$$

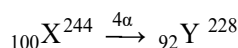
146. Ans (4)

$$\therefore \mu = \frac{c}{v} = \frac{3 \times 10^8}{2 \times 10^8} = \frac{3}{2}$$

From Brewster's law

$$\therefore i_p = \tan^{-1}(\mu) = \tan^{-1}\left(\frac{3}{2}\right)$$

147. Ans (3)



148. Ans (4)

First excited state $\Rightarrow n = 2$

$$T_2 = -13.6 \frac{z^2}{n^2} = -\frac{13.6}{4} \text{ eV}$$

Third excited state $\Rightarrow n = 4$

$$T_1 = -13.6 \frac{z^2}{n^2} = -\frac{13.6}{16} \text{ eV}$$

$$T_1 : T_2 = \frac{1}{16} : \frac{1}{4} = 1 : 4$$

149. Ans (3)

$$\text{Using, } V_0 = \frac{hv}{e} - \frac{hv_0}{e} \dots\dots(1)$$

$$\frac{V_0}{6} = \frac{hv/3}{e} - \frac{hv_0}{e} \dots\dots(2)$$

$$\text{On solving, } v_0 = \frac{v}{5}$$

150. Ans (3)

Diameter or radius $\propto n^2$

$$\frac{D}{1.06 \text{ Å}} = \left(\frac{5}{1}\right)^2$$

$$\Rightarrow D = 25 \times 1.06 \text{ Å}$$

$$= 26.50 \text{ Å}$$

151. Ans (2)

$$\text{Momentum of the incident photon } p = \frac{h}{\lambda},$$

$$\text{Momentum after reflection} = -\frac{h}{\lambda}$$

$$\text{Change in momentum} = \Delta p = \frac{2h}{\lambda}$$

If n is the number of photons falling per second on the screen then force

$$F = \frac{\Delta p}{\Delta t} = \frac{2nh}{\lambda}$$

$$\Rightarrow n = \frac{F\lambda}{2h} = \frac{1 \times 3300 \times 10^{-9} \times 10^{-6}}{2 \times 6.6 \times 10^{-34}}$$

$$= 2.5 \times 10^{21} \text{ photons s}^{-1}$$

152. Ans (1)

$$I_Z = \frac{P_Z}{V_Z} = \frac{0.27W}{9V} = 0.03A$$

$$I_L = \frac{V_L}{R_L} = \frac{9V}{450 \Omega} = 0.02A$$

$$I_S = I_Z + I_L = 0.05A$$

$$R_S = \frac{V_i - V_L}{I_S} = \frac{(15 - 9)V}{0.05 A}$$

$$R_S = 120 \Omega$$

153. Ans (3)

The P-N junction will conduct only when it is forward biased i.e. when + 5V is fed to it, so it will conduct for 2nd and 4th quarter part of signal shown and when it conducts potential drop 5 volt will be across both the resistors, so output voltage across R_2 is 2.5V.

$$\therefore V_0 = + 2.5V$$

154. Ans (4)

$$P_1 V_1 = P_2 V_2$$

$$30 \times 20 = P_2 \times 30$$

$$\text{Change in pressure} = 30 - 20 = 10 \text{ kPa}$$

155. Ans (1)

For constant heat flow rate (Q)

$$msdT = Qdt$$

$$\frac{dT}{dt} \propto \frac{1}{s}$$

For high specific heat, slope $\frac{dT}{dt}$ is lower that is for P.

156. Ans (4)

$$A_r = \frac{90}{100} \times A_i = \frac{9}{10} \times 0.01 = 0.009$$

No phase change at rarer medium.

$$\text{So } y = 0.009 \sin 10\pi \left[t + \frac{x}{10} \right]$$

157. Ans (2)

$$\Delta n = n_1 - n_2$$

$$= 130 - 125 = 5b/s$$

$$\frac{I_{\max}}{I_{\min}} = \left(\frac{a_1 + a_2}{a_1 - a_2} \right)^2 = \left(\frac{5+3}{5-3} \right)^2 = \frac{64}{4} = \frac{16}{1}$$

158. Ans (3)

$$\frac{1}{2}kA^2 = U_{\max} - U_{\min}$$

$$\Rightarrow \frac{1}{2}k \frac{100}{10^6} = 0.04$$

$$\Rightarrow k = 8 \times 10^2 \text{ N/m}$$

159. Ans (4)

$$\frac{v_1}{v_2} = \frac{16}{9} = \frac{r_1^2}{r_2^2}$$

$$\frac{r_1}{r_2} = \frac{4}{3}$$

$$\frac{V_1}{V_2} = \left(\frac{r_1}{r_2} \right)^3 = \frac{64}{27}$$

160. Ans (3)

Velocity of the rolling body when it reach the bottom

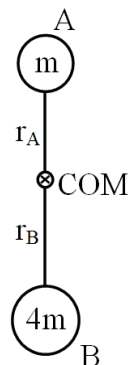
$$v = \sqrt{\frac{2gh}{1 + \frac{K^2}{R^2}}}$$

$$\therefore v \propto \left(\sqrt{1 + \frac{K^2}{R^2}} \right)^{-1}$$

$$\left(\frac{K^2}{R^2} \right)_{\text{ring}} = 1, \left(\frac{K^2}{R^2} \right)_{\text{disc}} = \frac{1}{2};$$

$$\left(\frac{K^2}{R^2} \right)_{\text{sphere}} = \frac{2}{5}, \left(\frac{K^2}{R^2} \right)_{\text{shell}} = \frac{2}{3}$$

161. Ans (3)



$$m_A r_A = m_B r_B \Rightarrow \frac{r_A}{r_B} = \frac{2m}{m} = 2$$

$$\because \omega_A = \omega_B$$

$$\frac{v_A}{r_A} = \frac{v_B}{r_B}$$

$$\frac{v_A}{v_B} = \frac{r_A}{r_B} = 2$$

162. Ans (2)

inelastic collision

$$\mu = 4mV$$

$$V = \frac{u}{4}$$

$$\text{COME } \frac{1}{2} 4m \left(\frac{u}{4} \right)^2 = 4mgh$$

$$h = \frac{u^2}{32g}$$

163. Ans (4)

$$f(x) = -\frac{dU}{dx}(x)$$

$$\text{or } U(x) = -\int F(x) dx$$

Here $F(x) = -kx$, where k is a positive constant.

$$U(x) = +k \int x dx = \frac{1}{2} kx^2$$

i.e. parabola

164. Ans (2)

$$\frac{b}{a} = \frac{T^{-1}}{ML^2 T^{-2}}$$

$$= M^{-1} L^{-2} T$$

165. Ans (4)

$$\text{Here } \frac{dv}{dx} = -1 \text{ and } v = 3 \text{ at } x = 2$$

$$\text{so } a = v \frac{dv}{dx} = 3(-1) = -3$$

$$\therefore \text{Magnitude of acceleration} = 3 \text{ m/s}^2$$

166. Ans (1)

$$a \propto t \Rightarrow V \propto t^2$$

$$a = \text{constant} \Rightarrow V \propto t$$

167. Ans (4)

Apply KVL from A to B

$$-20 + 2 \times 1 + 2 \times R + 2 \times 4 = 0$$

$$R = 5 \Omega$$

Hence option (4)

168. Ans (2)

$$\text{induced charge } q = \frac{\Delta\phi}{R} = \frac{16-4}{4} = \frac{12}{4} = 3.0 \text{ C}$$

169. Ans (1)

$$\frac{1}{f} = (\mu - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

$$\Rightarrow (1.7 - 1) \times \left[\frac{2}{21} \right]$$

$$f = 15 \text{ cm}$$

170. Ans (2)

$$\mu = \frac{1}{\sin \theta_c}$$

$$\frac{c}{v} = \frac{1}{\sin 37^\circ}$$

$$v = \frac{c}{(5/3)} = \frac{9 \times 10^8}{5} = 1.8 \times 10^8 \text{ m/s}$$

171. Ans (1)

NAND + NOT = AND

172. Ans (2)

$$\frac{\Delta A}{A} = \frac{2\Delta L}{L} = 8\%$$

173. Ans (3)

$$\lambda_m \propto \frac{1}{T}$$

$$\frac{\lambda_{m1}}{\lambda_{m2}} = \frac{T_2}{T_1}$$

$$\frac{\lambda_m}{\lambda_{m2}} = \frac{6000}{2000}$$

$$\lambda_{m2} = \frac{\lambda_m}{3}$$

174. Ans (2)

$$Y = \frac{\text{stress}}{\text{strain}} = \frac{10^9}{10^{-3}} = 10^{12} \text{ N/m}^2$$

175. Ans (1)

$$\frac{g}{16} = \frac{g}{\left(1 + \frac{h}{R}\right)^2}$$

$$4 = 1 + \frac{h}{R} \quad \boxed{h = 3R}$$

176. Ans (4)

$$R = 1 \text{ m}; m = 20 \text{ kg}; \omega = 40 \text{ rad s}^{-1}$$

$$\text{K.E.} = \frac{1}{2} I \omega^2 = \frac{1}{2} \left(\frac{mR^2}{2} \right) \omega^2$$

$$= \frac{1}{2} \times \left(\frac{20 \times 1}{2} \right) (40)^2 = 8000 \text{ J}$$

177. Ans (2)

$$T_{\max} = m\omega_{\max}^2 \cdot r$$

$$10 = \frac{1}{10} \times \omega_{\max}^2 \times 50 \times 10^{-2}$$

$$\omega_{\max} = 10\sqrt{2} \text{ rad/sec}$$

178. Ans (1)

$$\text{Pitch} = \frac{1 \text{ mm}}{2 \text{ Rotations}} = \frac{1 \text{ mm}}{2}$$

$$\text{LC} = \frac{\text{Pitch}}{100} = \frac{\frac{1}{2} \text{ mm}}{100} = 0.005 \text{ mm}$$

179. Ans (4)

$$a = \frac{9-3}{2+1} = 2 \text{ m/s}^2$$

$$\begin{array}{c} \xrightarrow{a} \\ \text{N} \rightarrow \boxed{1 \text{ kg}} \leftarrow 3\text{N} \end{array}$$

$$N - 3 = 1 \times a$$

$$N = 3 + 2 = 5\text{N}$$

180. Ans (2)

$$y = x - \frac{x^2}{9} = x \left[1 - \frac{x}{9} \right]$$

compare with

$$y = x \tan \theta \left[1 - \frac{x}{R} \right]$$

$$\boxed{R = 9\text{m}}$$